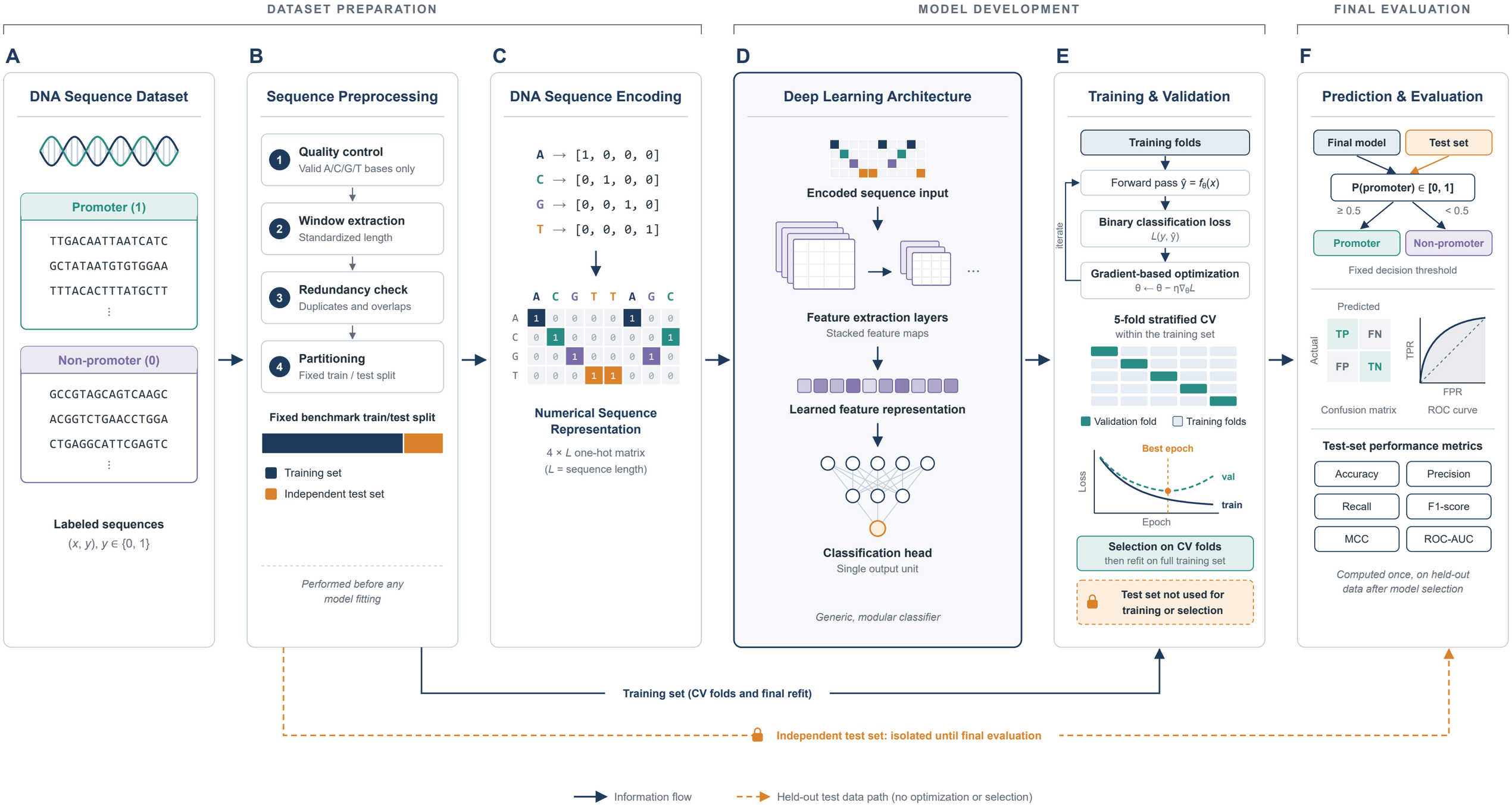

DATASET PREPARATION
MODEL DEVELOPMENT
FINAL EVALUATION
A
DNA Sequence Dataset
Promoter (1)
TTGACAATTAATCATC
GCTATAATGTGTGGAA
TTTACACTTTATGCTT
Non-promoter (0)
GCCGTAGCAGTCAAGC
ACGGTCTGAACCTGGA
CTGAGGCATTCGAGTC
Labeled sequences
(x, y), y ∈ {0, 1}
B
Sequence Preprocessing
1 Quality control
Valid A/C/G/T bases only
2 Window extraction
Standardized length
3 Redundancy check
Duplicates and overlaps
4 Partitioning
Fixed train / test split
Fixed benchmark train/test split
Training set
Independent test set
Performed before any model fitting
C
DNA Sequence Encoding
A → [1, 0, 0, 0]
C → [0, 1, 0, 0]
G → [0, 0, 1, 0]
T → [0, 0, 0, 1]
Numerical Sequence Representation
4 × L one-hot matrix
(L = sequence length)
D
Deep Learning Architecture
Encoded sequence input
Feature extraction layers
Stacked feature maps
Learned feature representation
Classification head
Single output unit
Generic, modular classifier
E
Training & Validation
Training folds
Forward pass ŷ = fθ(x)
Binary classification loss
L(y, ŷ)
Gradient-based optimization
θ ← θ − η∇θL
iterate
5-fold stratified CV
within the training set
Validation fold
Training folds
Best epoch
Loss
val
train
Epoch
Selection on CV folds
then refit on full training set
Test set not used for training or selection
F
Prediction & Evaluation
Final model
Test set
P(promoter) ∈ [0, 1]
≥ 0.5
< 0.5
Promoter
Non-promoter
Fixed decision threshold
Predicted
Actual
TP
FN
FP
TN
Confusion matrix
TPR
FPR
ROC curve
Test-set performance metrics
Accuracy
Precision
Recall
F1-score
MCC
ROC-AUC
Computed once, on held-out data after model selection
Training set (CV folds and final refit)
Independent test set: isolated until final evaluation
Information flow
Held-out test data path (no optimization or selection)